
A Patient World Model for Early Forecasting of Digital Health Campaign Outcomes: Capabilities and Limits

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Abstract

Digital direct-to-consumer (DTC) health campaigns are typically measured after they end. While current in-flight forecasts rely on standard classifiers with reusable samples, we replace this approach with a compact patient world model. Our architecture maintains a latent state per patient, learning exposure-conditioned dynamics alongside a weekly conversion hazard to smoothly roll forward into future conversion curves. Evaluated on a US campaign featuring 147,173 patients, the model predicted the final week-52 prescription volume from just week 4 with only a 2.9% error given recorded future exposures, whereas the strongest baseline erred by 13.6–33.1%. Crucially, when the future media plan is unknown, the model can imagine its own future exposures and roll out on them to yield equally accurate forecasts. A Fisher-information analysis confirms the necessity of this design. Without our dense next-exposure supervision, performance on rare outcomes severely degrades, driving the error rate from 2.9% up to 5.2–11.1%, even though common outcomes remain unaffected. Finally, we outline the model’s limitations, highlighting how strong selection effects prevent these observational rollouts from being interpreted causally.

1 Introduction

Digital direct-to-consumer (DTC) campaigns for prescription drugs produce patient-level exposure records. A patient may see campaign ads repeatedly across channels and ad formats over many weeks, while the outcome of interest, such as a specialist visit or a new prescription, may occur months later. Measurement today is mostly retrospective. Attribution and media-mix models [Shao and Li, 2011, Zhang et al., 2014, Jin et al., 2017] are fit only after outcomes have matured, but campaign teams cannot wait that long. Budgets, channel mix, and creative rotation must be adjusted mid-flight using partial data. The common in-flight practice is to train a separate classifier for every cutoff week τ and horizon H , what the forecasting literature calls a direct multi-horizon strategy [Ben Taieb et al., 2012]. It has two problems. First, nothing ties the classifiers together across horizons, so the assembled cumulative conversion curve can decrease, which is impossible for a true cumulative curve. Second, they contain no exposure-conditioned dynamics, so they cannot say what would happen under a different future plan.

The alternative is a recursive strategy: one model, rolled forward week by week. World models offer exactly that structure. They jointly learn how a latent state evolves under actions and what outcomes follow, then roll forward to simulate future trajectories [Sutton, 1991, Ha and Schmidhuber, 2018]. This framework now spans control [Hafner et al., 2020, 2025], interactive environments [Bruce et al., 2024], video simulation [Brooks et al., 2024, Assran et al., 2025], and driving [Hu et al., 2023], and is widely seen as the next frontier of machine intelligence [LeCun, 2022, Li, 2025].

Patient world models have begun to emerge as well. Delphi-2M [Shmatko et al., 2025] and ETHOS [Renc et al., 2024] generate plausible future health trajectories conditioned on patient history, but neither represents the intervention explicitly. Without that, a trajectory generator cannot answer the questions a decision maker actually asks: what happens under this plan, and what happens when the plan is unknown and the model must imagine it. Our setting supplies the missing piece. Exposures are recorded at patient-week granularity and outcomes are dated events, so forecasts can be evaluated against held-out observations at scale.

This paper develops and evaluates an intervention-conditioned patient world model. Our primary endpoint is new-to-brand prescriptions (NBRx), a rare outcome for which early forecasting is most valuable and hardest. Given the recorded future exposures, the model forecasts the final NBRx volume from a week-4 cutoff at the sampling floor of the test set, where common practice produces no usable forecast and a strong pooled-hazard baseline with the same rollout is off by several times more. The distinctive result is operational. In deployment the future media plan is unknown, so the world model imagines its own future exposure and rolls out on it, and it forecasts as accurately as when given the true future. Exposure is targeted rather than randomized, so every forecast in this paper is exposure-conditioned rather than causal, and one experiment quantifies the reversal that arises when these rollouts are read causally.

Our contributions are as follows.

1. **An architecture.** An intervention-conditioned patient world model comprising a swappable sequence-state module (here a GRU), a survival hazard head, and an auxiliary next-exposure prediction head. The components are trained jointly, and recursive survival rollout produces coherent conversion curves (Section 3).
2. **An analysis of where the gains arise.** We establish coherence and an additive bound on cumulative-incidence error in terms of weekly hazard errors, derive a concentration bound for aggregate volume under explicit assumptions, and use a Fisher-information argument to explain the potential value of dense auxiliary supervision in the rare-event regime (Section 4).
3. **Accurate early-cutoff forecasts of a rare outcome.** Given the future exposures, NBRx volume forecasts have relative errors of 0.8–2.9% across the evaluated cutoffs at weeks 4–26, against 13.6–33.1% for the pooled-hazard gradient boosting baseline. Removing the auxiliary objective raises volume error to 5.2–11.1% on this outcome and on no other, supporting the predicted regime dependence (Section 6).
4. **Operational forecasting by imagination.** When the future media plan is unknown, the world model imagines its own future exposure and rolls out on it, forecasting as accurately as the same model handed the true future and more accurately than a strong survival baseline that is given the true future (Section 6.3). This is a capability that per-horizon classifiers and outcome-only sequence models do not have.
5. **The limits, stated.** On the denser diagnosis outcome the imagined rollout degrades. Turning all future exposure off raises predicted conversion from 0.31 to 0.89 in our model, with a similar reversal in the baseline simulator. This is a selection effect, not a response, and it shows why these observational rollouts cannot be read causally (Section 6.4).

2 Related work

World models. Learning environment dynamics together with the outcome signal, then rolling the model forward, is the core pattern of model-based reinforcement learning [Sutton, 1991, Ha and Schmidhuber, 2018, Hafner et al., 2020, 2025]. The same recipe has scaled to generative interactive environments [Bruce et al., 2024], video models used as general-purpose simulators [Brooks et al., 2024, Assran et al., 2025], and driving [Hu et al., 2023]. We borrow the structure: latent state, action-conditioned transition, joint training, and rollout. We do not borrow the control objective. No policy is optimized, and our actions are observed exposures. The structure buys us measurement: coherent forecasts at every horizon, plus scenario simulation under stated assumptions.

World models for health. Generative models of patient trajectories are emerging fast. Delphi-2M models the natural history of over 1,000 diseases from biobank-scale records [Shmatko et al., 2025]. ETHOS predicts tokenized patient timelines zero-shot [Renc et al., 2024]. MOTOR pre-trains a time-to-event foundation model on structured medical records [Steinberg et al., 2024], and

digital-twin models generate synthetic control arms for trials [Fisher et al., 2019]. Surveys note that most EHR foundation models are still evaluated far from decisions [Wornow et al., 2023]. All of these model the outcome process. The intervention process is absent or folded implicitly into the history. Our model is far smaller, but its action stream is explicit and weekly. That is what makes intervention-conditioned rollout and scenario simulation well-posed tasks. We view the two lines as complementary. Trajectory foundation models supply rich patient states. An explicit intervention-conditioned transition turns a trajectory generator into a decision tool.

Treatment-response forecasting over time. Recurrent marginal structural networks [Lim et al., 2018] and counterfactual recurrent networks [Bica et al., 2020] estimate outcomes under hypothetical treatment sequences. They inherit the assumptions of marginal structural models, such as sequential ignorability [Robins et al., 2000]. We claim less. Our model produces exposure-conditioned forecasts of observed outcomes, and scenario rollouts are simulations under stated assumptions. Section 6.4 shows concretely what goes wrong when this distinction is ignored.

3 An intervention-conditioned patient world model

3.1 Setup and tasks

Consider a campaign observed week by week. For patient i , let $r \in \{0, 1, \dots, R\}$ index the week relative to enrollment, with $R = 52$ in our data. During week r the patient receives digital exposure summarized by an *action* vector $\mathbf{a}_{i,r} \in \mathbb{R}^{d_a}$. It collects log-transformed impression counts by channel, ad type, and targeting tactic. A static vector $\mathbf{x}_i \in \mathbb{R}^{d_s}$ holds eligibility flags, source of business, and baseline activity. The word action means recorded observational exposure. Delivery is targeted, not randomized. Every quantity in this paper is therefore an exposure-conditioned forecast, not a causal effect. Section 6.4 makes this concrete.

Let T_i be the week of the conversion event and let C_i be the censoring week, set by the end of the observation window. Following the discrete-time survival convention [Cox, 1972, Tutz and Schmid, 2016], define the per-week hazard

$$h_{i,r+1} = \mathbb{P}(T_i = r+1 \mid T_i > r, \mathbf{x}_i, \mathbf{a}_{i,1:r}), \quad (1)$$

the probability that a patient at risk through week r converts in week $r+1$. A patient contributes one at-risk observation for every week $r < \min(T_i, C_i)$. The cumulative incidence through week k is

$$F_i(k) = 1 - \prod_{m=1}^k (1 - h_{i,m}), \quad (2)$$

which never decreases in k as long as the hazards are valid probabilities. Neural survival models adopt the same convention [Katzman et al., 2018, Lee et al., 2018, Giunchiglia et al., 2018], but they treat the covariate process as given. That choice supports risk scoring and rules out action-conditioned simulation.

Three tasks are studied. **(T1) One-step hazard prediction.** Given the history through week r , predict $h_{i,r+1}$. This is a foundation check. A model that cannot learn the one-step signal has nothing to roll out. **(T2) Early trajectory forecasting.** Given only the history through a cutoff $\tau < R$, forecast the future curve $F_i(k)$ for every patient still at risk, then aggregate to the campaign volume forecast

$$\hat{N}(k) = \sum_{i \in \mathcal{R}_\tau} w_i \frac{F_i(k) - F_i(\tau)}{1 - F_i(\tau)}. \quad (3)$$

Here $\mathcal{R}_\tau$ is the at-risk set at τ . The weights w_i are inverse-sampling weights [Horvitz and Thompson, 1952] that map the modeling cohort back to natural prevalence. **(T3) Scenario simulation.** From cutoff τ , replace the future exposures with a hypothetical plan $\mathbf{a}'_{i,\tau+1:R}$ and recompute (2). The stated assumption is that the learned dynamics hold on the support of the observed data. Task (T2) is where per-horizon practice breaks down, since its per-cell probabilities share no state and need not be monotone. Per-cutoff models are close to landmarking in survival analysis [van Houwelingen, 2007], except that landmarking keeps the survival likelihood and the per-horizon classifiers drop it. Task (T3) additionally requires an explicit exposure-conditioned transition model, which neither per-horizon classifiers nor outcome-only sequence models have.

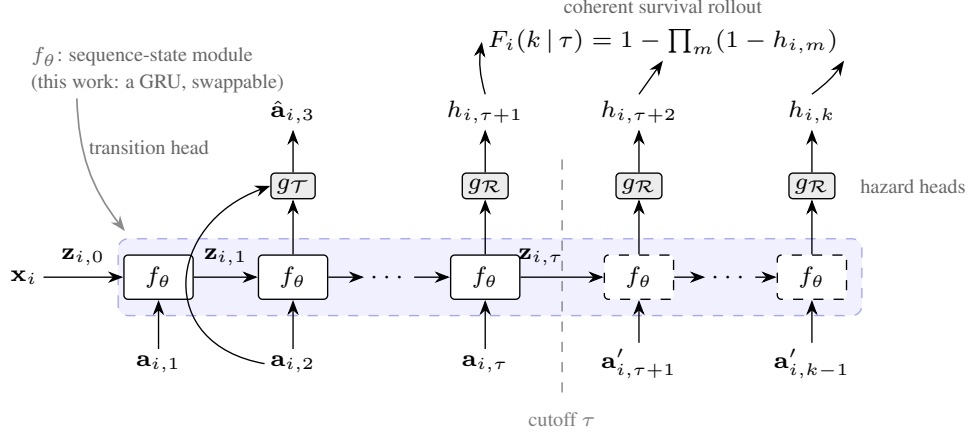


Figure 1: The proposed architecture, unrolled. It is defined by three components and one inference pattern: a swappable sequence-state module f_θ (instantiated as a GRU in this work), a hazard head g_R , a transition head g_T for dense next-action supervision, and the recursive survival rollout that multiplies weekly hazards into a coherent cumulative-incidence forecast. The transition head takes $(z_{i,r}, a_{i,r})$, shown by the curved input. During training both heads apply at every at-risk week. The figure shows g_T once on the observed prefix and g_R from the cutoff onward, where forecasting happens. After the cutoff (dashed cells), the same module runs on recorded or hypothetical exposures a' .

3.2 Architecture

The architecture is defined by three components and one inference pattern (Fig. 1): a sequence-state module f_θ , a hazard head g_R , a transition head g_T , and a recursive survival rollout. The sequence module is deliberately generic. We instantiate it as a two-layer GRU [Cho et al., 2014] with hidden size 128 and about 2×10^5 parameters, which trains on CPU inside the certified environments where patient data must stay (Appendix C). Any sequence model can take its place, and a stronger module may improve the results. The GRU is not the contribution. In fact, a plain GRU forecaster is one of our baselines. It is obtained by deleting one component, and Section 6 measures exactly what that deletion costs.

The static vector sets the initial state through a learned projection, and the module advances the state with each week’s action,

$$z_{i,0} = \tanh(W_0 x_i + b_0), \quad z_{i,r} = f_\theta(z_{i,r-1}, a_{i,r}), \quad (4)$$

so that $z_{i,r}$ summarizes what the campaign has done to patient i in the first r weeks. The hazard head is a small two-layer network that outputs a hazard at every at-risk week,

$$h_{i,r+1} = \sigma(g_R(z_{i,r})). \quad (5)$$

The transition head predicts the next week’s action summary from the current state and action,

$$\hat{a}_{i,r+1} = g_T(z_{i,r}, a_{i,r}). \quad (6)$$

The target of (6) is the observable exposure summary itself. Transition quality can therefore be checked directly on held-out data rather than taken on faith.

3.3 Training objective

Let $\mathcal{A}$ be the set of at-risk patient-weeks (i, r) in the training split, and let $y_{i,r+1} \in \{0, 1\}$ indicate a conversion in week $r+1$. The model is trained end-to-end by minimizing

$$\mathcal{L} = - \sum_{(i,r) \in \mathcal{A}} \left[y_{i,r+1} \log h_{i,r+1} + (1-y_{i,r+1}) \log(1-h_{i,r+1}) \right] + \lambda \sum_{(i,r) \in \mathcal{A}'} \|\hat{a}_{i,r+1} - a_{i,r+1}\|_2^2, \quad (7)$$

where $\mathcal{A}'$ keeps the patient-weeks whose next week is observed, and $\lambda = 0.3$ in all experiments. The first term is the discrete-time survival likelihood over pooled at-risk person-weeks [Tutz and Schmid,

2016]. The second term is the dynamics supervision. Setting $\lambda = 0$ and dropping (6) recovers exactly a GRU forecaster of the same capacity. The comparison between the two in Section 6 is therefore a one-flag ablation, not a reimplementaion.

3.4 Rollout and forecasting

Given a cutoff τ , the module reads the observed prefix and produces $\mathbf{z}_{i,\tau}$. The model is then unrolled. The hazard for week $\tau+1$ reads $\mathbf{z}_{i,\tau}$ directly. For each later week m , the recurrence (4) is first advanced with the action $\mathbf{a}_{i,m-1}$ to form $\mathbf{z}_{i,m-1}$, and the hazard head (5) then outputs $h_{i,m}$. The forecast cumulative incidence from the cutoff is

$$F_i(k | \tau) = 1 - \prod_{m=\tau+1}^k (1 - h_{i,m}), \quad k \in (\tau, R], \quad (8)$$

monotone and consistent across every horizon by construction. This cutoff-conditional incidence is exactly the summand $[F_i(k) - F_i(\tau)]/[1 - F_i(\tau)]$ of the volume aggregation (3), which therefore takes the rollout output directly, with no further conditioning. The future actions can be the recorded exposures, a stated continuation rule, or a hypothetical scenario $\mathbf{a}'$.

They can also be supplied by the model itself. When the future plan is unknown, the transition head (6) generates each next exposure and the rollout consumes its own output,

$$\hat{\mathbf{a}}_{i,m} = g_{\mathcal{T}}(\mathbf{z}_{i,m-1}, \mathbf{a}_{i,m-1}), \quad \mathbf{z}_{i,m} = f_{\theta}(\mathbf{z}_{i,m-1}, \hat{\mathbf{a}}_{i,m}), \quad m > \tau, \quad (9)$$

seeded at the cutoff by the last observed action and closed with no access to anything after week τ . This is the world model rolling forward in its own imagination, and Section 6.3 tests it.

Each task exercises one component. Task (T1) uses the state and the hazard head. Task (T2) adds the rollout over recorded exposures. Task (T3) adds counterfactual exposure substitution, replacing the recorded future actions with a hypothetical plan or with the model’s own imagined plan from (9).

4 Analysis: what the structure predicts

This section analyzes the formulation, not the sequence module. Every statement below holds for any encoder f_{θ} . Proposition 1 is a property of the survival rollout. Proposition 2 is a property of the aggregation. The Fisher argument is a property of the joint objective (7). Proofs are in Appendix G. Throughout, $\{h_{i,m}\}$ are the true hazards and $\{\hat{h}_{i,m}\}$ are the model’s estimates.

4.1 Error propagation through the rollout

One may worry that rollout errors compound multiplicatively with the horizon. The product form (8) rules this out.

Proposition 1 (Coherence and error propagation) *Fix any estimates $\hat{h}_{i,m} \in [0, 1]$ and let $\hat{F}_i(k|\tau)$ be computed by (8). Then (i) $\hat{F}_i(k|\tau)$ is non-decreasing in k and $[0, 1]$ -valued for every realization, and (ii)*

$$|\hat{F}_i(k|\tau) - F_i(k|\tau)| \leq \sum_{m=\tau+1}^k |\hat{h}_{i,m} - h_{i,m}|, \quad \forall k \in (\tau, R]. \quad (10)$$

The trajectory error grows at most additively with the horizon, in increments that are differences of small numbers. Weekly hazards average $\bar{h} \approx 9 \times 10^{-3}$ for the diagnosis outcome in our data, and an order of magnitude less for the rarer prescription outcome. Part (i) costs nothing. It holds for any hazard estimates and for any model given the pooled-hazard-plus-rollout form, boosted trees included. The per-horizon practice has neither property. Each (τ, H) classifier estimates $F_i(k)$ on its own, so monotonicity can fail. A plain classifier also has no way to use censored patients. Excluding them selects converters into the long-horizon training slices, a bias that Section 6.1 measures at full size.

4.2 Concentration of the campaign-volume forecast

The next question is when the aggregate forecast (3) can be trusted. Condition on the cutoff-time information $\mathcal{F}_\tau$. For each patient $i \in \mathcal{R}_\tau$, let

$$q_i = \frac{\hat{F}_i(R|\tau) - \hat{F}_i(\tau)}{1 - \hat{F}_i(\tau)}$$

be the predicted probability of converting in the remaining weeks, and let $N = \sum_{i \in \mathcal{R}_\tau} w_i \mathbf{1}\{\tau < T_i \leq R\}$ be the realized weighted converter count.

Proposition 2 (Volume concentration) *Assume (i) conversions of distinct patients are independent conditional on $\mathcal{F}_\tau$, and (ii) the model is trajectory-calibrated on the risk set, i.e., $\mathbb{P}(\tau < T_i \leq R | \mathcal{F}_\tau) = q_i$. Then $\mathbb{E}[N | \mathcal{F}_\tau] = \hat{N}(R)$, and for all $t > 0$,*

$$\mathbb{P}\left(|N - \hat{N}(R)| \geq t \mid \mathcal{F}_\tau\right) \leq 2 \exp\left(-\frac{t^2/2}{V + \frac{1}{3}w_{\max}t}\right), \quad (11)$$

where $V = \sum_{i \in \mathcal{R}_\tau} w_i^2 q_i (1 - q_i)$ and $w_{\max} = \max_i w_i$.

Proposition 2 splits the volume error into a random part and a systematic part. The random part is bounded by (11). The systematic part is the calibration bias that breaks assumption (ii). Three consequences are used in the experiments. Single-digit-percent volume errors are calibration bias, not sampling noise. Per-cell errors of hundreds of percent cannot be noise at this sample size, so they are structural. And once two coherent models are both well calibrated, the campaign-level aggregate stops separating them, even when patient-level discrimination still does. The 1σ relative floor is about $1/\sqrt{n_+}$, with n_+ the expected converter count. Every volume forecast below is read against this floor.

4.3 Dense supervision in the rare-event regime

The last question is why the dense next-exposure supervision should help, and when. A Fisher-information comparison of the two loss terms, given in Appendix H, has a simple reading. The hazard term’s curvature carries a factor $h(1 - h)$, small precisely because conversions are rare, while the transition term carries no such factor and contributes at every observed week across all d_a exposure components. When events are rare, the next-state loss therefore supplies most of the curvature that pins down the shared representation. This is a scaling argument rather than an exact ratio, and it makes a testable prediction: the transition supervision should help most in the rare-event regime.

4.4 Three testable predictions

P1 (structure). Coherence and additive error propagation belong to the rollout, not the network. Any pooled-hazard model with the rollout, including a plain GBM, should beat the per- (τ, H) practice by a wide margin.

P2 (aggregate ceiling). Once coherent models are well calibrated, their volume errors should cluster near the sampling floor. Aggregates then stop separating architectures, while patient-level discrimination still can.

P3 (regime dependence). The value of the learned state and of the dense transition supervision should grow as the outcome gets rarer and as the sequences carry more dynamics. On the common diagnosis outcome the world model should be close to a strong pooled-hazard baseline. On the rare prescription outcome it should pull far ahead.

The headline experiment sits exactly where P3 predicts the largest gap. The supporting results for P1 and P2, where the world model is predicted not to dominate and indeed does not, are in Appendix D.

5 Experimental setup

Data. The data come from a large US digital DTC campaign for a prescription cardiovascular drug. The records are protected health information. They stay on a secured analytics cluster and only

Table 1: The headline task: forecast the campaign’s final NBRx volume from an early cutoff τ (rx outcome, test split, known future exposures). Relative error (%) of the forecast final converter count. “Coherent” is the fraction of monotone per-patient curves (range across cutoffs). Test risk sets hold 21,963 down to 20,595 patients. Observed future converters $n_+ = 711, 601, 489, 206$.

Model	NBRx volume rel. error (%) at cutoff τ				Coherent
	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$	
Per- (τ, H) GBM	2582.0	3054.8	3754.7	8491.6	19–24%
Per- (τ, H) GBM + future	2581.8	3054.1	3754.0	8491.9	11–31%
Naive extrapolation	64.4	82.6	97.4	166.1	100%
Pooled-hazard GBM	13.6	17.3	19.2	33.1	100%
GRU forecaster ($\lambda=0$)	5.2	7.4	8.2	11.1	100%
GRU world model (proposed)	2.9	2.3	2.6	0.8	100%

aggregate metrics are reported. From a universe of 10.9M eligible patients, a stratified case-control cohort of 147,173 patients is built and split 70/15/15 by patient hash. The pooled at-risk person-week count is 5.2M. The weekly action vector ($d_a = 17$) holds winsorized log impression counts in total and by channel, ad-type, and targeting buckets, an activity flag, and the normalized relative week. The static vector ($d_s = 7$) holds source-of-business indicators, eligibility flags, baseline visit volume, and the enrollment week. Two outcomes are studied. The diagnosis outcome (dx, a visit to a relevant specialist) has about 49k events in the cohort. The prescription outcome (rx, a new-to-brand prescription) has 5,225 events and is roughly ten times rarer. By prediction P3, rx is where the world model should matter most. Two design facts matter when reading the results. The cohort is enriched to 33.6% dx prevalence on purpose, and every metric is reported both on the enriched cohort and reweighted to natural prevalence with the inverse-sampling weights w_i . The sequences are sparse, with a median of three active weeks per patient. Cohort construction and full feature definitions are in Appendices A and B.

Baselines, protocol, and metrics. Every model sees the same features, splits, and test patients. The trajectory baselines are: **per- (τ, H) GBM** and **GBM+future**, LightGBM [Ke et al., 2017] trained separately for each cutoff-horizon cell, reflecting common practice, with the +future variant also given aggregated future exposures. **Pooled-hazard GBM**, a single discrete-time pooled-hazard LightGBM rolled out exactly like our model. It is one coherent model for all horizons and our strongest baseline. **GRU forecaster**, the $\lambda = 0$ ablation. **Naive**, which extrapolates the campaign-to-date constant hazard. The one-step task additionally uses logistic regression, flat LightGBM, and an MLP (Appendix D). Campaign-level metrics per cutoff are the relative *volume error* of the final converter count (3) against the Kaplan–Meier [Kaplan and Meier, 1958] count on the risk set, the *trajectory calibration* (ICI, the mean absolute gap between predicted and observed curves), and *coherence* (the fraction of monotone per-patient curves). Patient-level metrics are AUROC, AUPRC, and calibration, both enriched and IPW-corrected. The transition head is scored by held-out next-state skill against mean and persistence predictors, so the claim that it learns dynamics is checked rather than asserted.

6 Results

The results build to one capability. We first show that, given the recorded future exposures, the world model calls the year-end volume of a rare outcome at the sampling floor (Section 6.1), then locate where that accuracy comes from (Section 6.2). The payoff is operational (Section 6.3): with the future plan unknown, the model imagines it and forecasts just as well, which is something no per-horizon or outcome-only method can do.

6.1 Forecasting the year-end outcome by rollout from an early cutoff

The task: after observing the first τ weeks, forecast the number of new-to-brand prescriptions that the at-risk audience will generate by week 52. The evaluation feeds every model the recorded future exposures, so the comparison isolates learned response dynamics. Forecasts are per-patient curves (8) aggregated by (3) and scored against the Kaplan–Meier count on the same risk set. Table 1 is

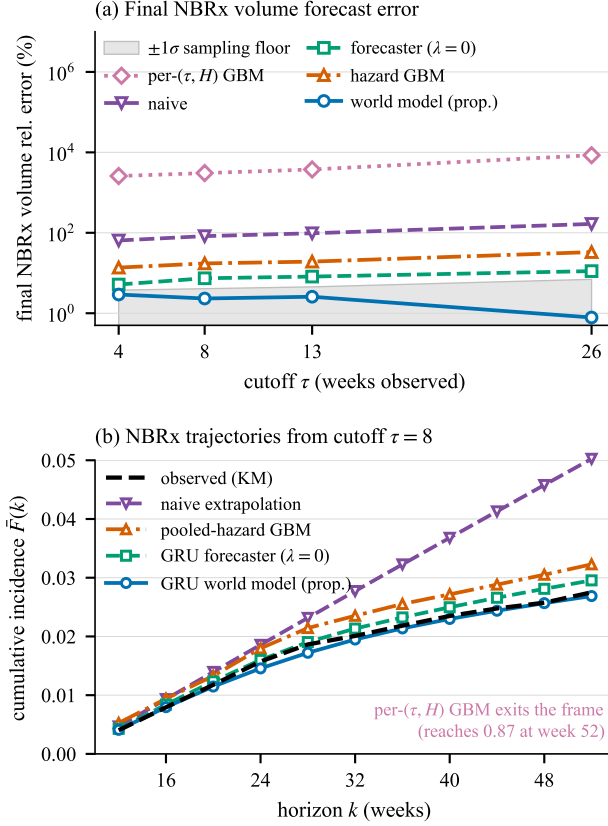


Figure 2: The headline result (rx outcome, test split). (a) Relative error of the forecast final NBRx volume versus the cutoff τ (log scale). The shaded band is the $\pm 1\sigma$ sampling floor implied by Proposition 2 ($\approx 1/\sqrt{n_+}$). No method, however good, can be expected to beat it. The world model sits inside the floor at every cutoff. The fair coherent baseline sits 4–5 σ above it, and the per-horizon practice two to four orders of magnitude above it. Read horizontally, a small τ means a long rollout ($\tau = 4$ rolls 48 weeks), and the world model’s error does not grow with rollout length, as Proposition 1 requires. (b) Predicted NBRx cumulative-incidence trajectories from cutoff $\tau = 8$ against the observed curve, estimated by Kaplan–Meier (KM), the standard estimator of cumulative incidence under right-censoring. The world model tracks the observed curve across all 44 remaining weeks. The whole curve is calibrated, not just the endpoint. The per-cell curve leaves the frame.

the exact record. Fig. 2 plots the same numbers against the sampling floor. At $\tau = 4$, with about 711 future converters in a risk set of 21,963 test patients, the world model forecasts 691 against 711 observed, an error of 2.9%. From $\tau = 26$ it forecasts 208 against 206 (0.8%). Converter counts are inverse-sampling-weighted estimates rounded for display, and the relative errors are computed on the unrounded values. Every curve is monotone by construction.

How close is that to the best possible? By Proposition 2, even a perfectly calibrated forecaster faces a sampling floor of about $1/\sqrt{n_+}$, because the realized converter count is itself random. On these risk sets the 1σ floor is 3.7%, 4.1%, 4.5%, and 7.0% at $\tau = 4$ –26. The world model’s errors of 2.9, 2.3, 2.6, and 0.8% sit at or below the floor at every one of these cutoffs. On this test set it is statistically indistinguishable from a perfectly calibrated forecaster. There is no headroom left to measure. The pooled-hazard GBM sits 3.6–4.7 σ above the floor at the same cutoffs, which is systematic bias rather than noise.

The per-horizon practice does not produce a usable volume forecast here, for a structural reason rather than a matter of tuning. A per-cell classifier can label a non-converter at horizon k only if that patient is observed through week k . Its long-horizon training slices therefore keep every converter

but only the earliest-enrolled non-converters. The slices become converter-dominated, the classifier learns a conversion rate near 0.9, and it applies that rate to a risk set whose true rate is 0.03. The volume error is 2,582% at week 4 and grows with the horizon (Table 1). Three signatures mark this as the known failure of horizon-wise labeling under right-censoring [Tutz and Schmid, 2016], not a coding error. It scales with rarity, from 209% on the ten-times-more-common dx outcome to thousands of percent here (Appendix D), exactly as a selection-through-censoring bias must. It is unmoved by information, since the variant given the aggregated future exposures errs by the same 2,582%. And on the rare outcome the bias collapses even patient ranking, leaving the per-cell IPW-corrected AUROC near chance (0.51–0.58). Fewer than a third of the assembled curves are monotone.

The remedy is the survival formulation, not a better classifier. The same LightGBM, given the survival rollout and the same features and future exposures, becomes the pooled-hazard GBM, our strongest baseline. Structure alone carries it to 13.6–33.1%, far beyond common practice, yet 4.7 to 41 times the world model’s error across cutoffs. The world model’s trajectory ICI is also three to six times tighter, and both GRU models add about +0.02 AUROC over the GBM baseline (Appendix D). Read vertically, Fig. 2(a) is an ablation ladder. At the week-4 cutoff, coupling the horizons at all is worth a factor of 40, the survival rollout another 5, and the learned state another 3. The transition supervision takes the last step, from 5.2–11.1% down to 0.8–2.9%, dissected next. Panel (b) shows that the whole curve is calibrated, so one trained model answers every interim milestone question with the same accuracy.

One qualification carries over to the rest of the paper. This is a retrospective evaluation with known future exposures, applied equally to every model. Section 6.3 takes the step to a fully operational forecast, where the future plan is not known.

6.2 Where the learned dynamics pay off

The headline number would be suspicious if the world model won everywhere. Strong methods usually have a regime, and the analysis of Section 4 says where this one should be. Two checks confirm the mechanism.

First, the gap appears exactly where P3 puts it. On the diagnosis outcome, with ten times more events on the same patients and pipeline, the three coherent models land in the same single-digit band. The world model reaches 4.5–7.5% for $\tau \leq 26$ and the GBM baseline 6.0–9.2%, which is the aggregate ceiling of P2. The per-cell practice still fails by 209–645%, as P1 predicts. The full dx tables, the one-step task, and a dynamics-rich subgroup analysis are in Appendix D. The decisive advantage is confined to the rare outcome. That is what the Fisher argument requires. With $\bar{h}$ an order of magnitude smaller, the outcome likelihood alone leaves the encoder loosely determined, so the models that share the dense supervision should separate from those that do not.

Second, the one-flag ablation isolates the ingredient. Setting $\lambda = 0$ removes exactly one thing, the transition loss. On dx the ablation costs nothing, and forecaster and world model are statistically inseparable there. On rx it raises volume error at every cutoff, from 2.9/2.3/2.6/0.8% to 5.2/7.4/8.2/11.1% at $\tau = 4$ –26, with discrimination essentially tied. One flag, one direction, only in the rare-event regime, as the Fisher argument of (12)–(13) predicts. The location and the size of the effect are hard to explain by anything except the dense supervision.

6.3 Forecasting by imagination: closed-loop rollout without a known future

Every result so far handed each model the recorded future exposures. That is a fair comparison but not a deployable one, because in operation no one knows the future media plan. This is where the world model does something the other methods cannot. It imagines its own future exposure with the transition head (6) and rolls out on the imagined sequence by (9), seeing nothing after the cutoff. The transition head predicts next week’s exposure with +11.5% held-out skill over a mean predictor (Appendix C). The plan it imagines is therefore not a guess, though it is not the true media plan either. Closed-loop rollouts are known to compound their own errors [Ross et al., 2011, Bengio et al., 2015, Janner et al., 2019], so this is the harder test.

Every baseline instead needs an assumed future plan, and we supply three: all future exposure off, the last observed week held constant, and the train-cohort average plan. The assumption is load-

Table 2: Operational forecasting (rx outcome, test split): forecast the final NBRx volume from cutoff τ without access to the future media plan. Relative error (%) of the forecast final converter count. The world model imagines its own future exposure and rolls out closed loop. Baselines are handed an assumed future plan, or, in the two reference rows, the true recorded future. All per-patient curves are coherent.

Method	Future source	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$
GRU world model	imagined (closed loop)	2.5	1.7	0.5	1.8
GRU world model	true future (ref.)	2.9	2.3	2.6	0.8
GRU world model	mean plan	2.4	3.6	1.2	3.3
GRU forecaster ($\lambda=0$)	mean plan	64.2	53.5	38.6	20.2
Pooled-hazard GBM	true future (ref.)	13.6	17.3	19.2	33.1
Pooled-hazard GBM	mean plan	37.5	39.2	38.7	52.3
Naive extrapolation	—	64.4	82.6	97.4	166.1

Table 3: Scenario simulation (T3): simulated dose-response on the dx outcome. Each cell is the mean predicted cumulative conversion of the test risk set when all future exposure is scaled by α ($do(\alpha \cdot \mathbf{a})$), for the world-model simulator and the perturbed pooled-hazard GBM. The $\alpha = 0$ column is a selection artifact, not an effect (see text). The $\alpha = 4$ column is outside the observed dose support.

Simulator (τ)	Dose scale α						
	0	0.25	0.5	1	1.5	2	4
World model ($\tau=4$)	0.893	0.312	0.310	0.310	0.310	0.312	0.316
Pooled-hazard GBM ($\tau=4$)	0.632	0.315	0.314	0.313	0.318	0.325	0.364
World model ($\tau=8$)	0.708	0.267	0.265	0.265	0.266	0.266	0.270
Pooled-hazard GBM ($\tau=8$)	0.504	0.273	0.272	0.271	0.275	0.279	0.307

bearing. Assuming no further exposure is the off-support scenario that Section 6.4 shows inflates predicted conversion. Implementation details of the imagined rollout are in Appendix E.

Table 2 reports the rx result. The closed-loop world model, which sees none of the future, forecasts the final NBRx volume within 0.5 to 2.5% at every cutoff. This matches the same model handed the true future exposures, 0.8 to 2.9% in Table 1. Both sit inside the sampling floor of 3.7 to 7.0%, so the two are statistically indistinguishable. Imagining the future plan costs nothing here. The trajectory calibration is also unchanged, with a closed-loop ICI of 0.0005 to 0.0008 that matches the known-future values. Against the baselines the gap is wide. Given the train-cohort mean plan, the pooled-hazard GBM errs by 38 to 52%, and no closed-book plan brings it below 35% at any cutoff (Appendix E). Even fed the true future plan, it errs by 13.6 to 33.1%. A model that imagines its own future is more accurate than a strong survival model that is handed the real one.

Two controls locate the source of this accuracy. First, hold the plan fixed and vary only the training objective. Under the same mean plan the world model forecasts within 1.2 to 3.6%, while the plain GRU forecaster ($\lambda = 0$) errs by 20 to 64%. The transition objective, not the imagination step by itself, is what makes the learned state robust to an imperfect plan. Second, export the imagined exposures and feed them to the GBM hazard head. This hybrid errs by 59 to 77%, worse than the GBM’s own mean plan. The imagined exposures are not a useful feature in isolation. The value lives in the world model’s latent state consuming its own rollout. The imagined plan need not be point-accurate as data. It only needs to be good enough for the same model’s state to roll forward on, which is the sense in which this is a world model and not a synthetic-data generator.

This capability is specific to the rare outcome. On the ten-times-denser dx outcome the closed-loop rollout degrades to 15 to 27%, and there the coherent baselines are more accurate. The likely cause is that the model reacts to whether exposure is present rather than to how much, and an imagined plan blurs that distinction. Appendix E gives the full comparison for both outcomes and the evidence on the mechanism.

6.4 Counterfactual simulation and a selection artifact

The transition model also turns the forecaster into a simulator. Replace the future exposures with a stated scenario $\mathbf{a}'$, re-roll, and read off the predicted trajectory. We exercised this with dose scaling ($\mathbf{a}' = \alpha \mathbf{a}$, $\alpha \in [0, 4]$), per-lever shutoffs, and a placebo control, on dx at $\tau \in \{4, 8\}$ (protocol in Appendix F). Within the observed dose support the simulator behaves sanely (Table 3). The dose-response is small, about +0.6 percentage points for a $4\times$ dose, and essentially flat within the observed support. The placebo moves the forecast by less than 10^{-4} , and no single lever moves it by more than 0.003.

The instructive result is off support. Setting $\alpha = 0$ switches all future exposure off, and on a causal reading this should produce the lowest conversion. It produces the highest by far, jumping from 0.31 to 0.89. The mechanism is selection through censoring: a patient’s exposure stops once they convert, so no future exposure is itself strong evidence of imminent conversion. The GBM baseline simulator shows the same inversion at 0.63, so this is a property of observational exposure data under outcome-dependent censoring, not a defect of one architecture. Formally, a scenario rollout is a g-computation [Robins, 1986] without the identification assumptions that would make it causal. Turning scenario outputs into effect estimates requires design, such as holdouts, randomization, or explicit deconfounding [Robins et al., 2000, Künzel et al., 2019].

7 Conclusion

We formulated in-flight campaign measurement as exposure-conditioned patient-state trajectory modeling and built a compact patient world model for it. From four weeks of observation, the model calls the campaign’s final volume on the rare business endpoint within 2.9%, at the sampling floor of the test set, a task on which standard practice cannot produce a usable forecast. The operational result is the one we would highlight. With the future media plan unknown, the model imagines its own exposure and forecasts just as accurately as when handed the true future, and more accurately than a strong survival baseline that is handed the true future. The mechanism is understood rather than guessed. A Fisher-information argument locates the value of dense transition supervision in the rare-event regime, and a one-flag ablation confirms both the location and the size of the effect. The same recipe, a survival-coherent rollout from a learned intervention-conditioned state that can imagine its own actions, applies wherever longitudinal exposures meet sparse terminal outcomes. Adherence programs, vaccination campaigns, and public-health communication are natural next targets.

Responsible use and limitations

This work forecasts observed health-seeking behavior under recorded digital exposure. It does not estimate causal effects, and Section 6.4 quantifies why its scenario outputs must not be read as such. Model outputs are decision support for campaign measurement and planning. They are not used, and should not be used, for individual-level clinical or coverage decisions. Uncertainty is reported throughout against the sampling floor of Proposition 2. All results come from one campaign, one therapeutic area, and one country, on an intentionally enriched cohort that is reweighted to natural prevalence at evaluation. External validity is untested. The retrospective evaluations condition on recorded future exposures, applied equally to every model. The operational closed-loop forecast (Section 6.3) removes that assumption on the rare rx outcome, where it matches the known-future forecast. On the denser dx outcome the closed-loop rollout degrades (Appendix E), so this capability is established for rare outcomes and not yet for dense ones. The underlying data are protected health information analyzed inside a certified environment. Only aggregate metrics are reported, and no patient-level output leaves that environment. Extended limitations and deployment guidance are in Appendices D and I.

Reproducibility and data availability. The underlying data cannot be shared. Cohort construction rules, feature definitions, hyperparameters, and the full result tables are given in Appendices A–I, at a level of detail sufficient to rebuild the pipeline on comparable data.

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A Cohort construction

The modeling universe contains 10.88M patients with campaign eligibility. The stratified case-control cohort keeps: all rx converters (5,225), all on-market comparator patients (4,399), a deterministic-hash subsample of dx converters (44,917), and a deterministic-hash subsample of on-market non-converting negatives with at least eight observed weeks (95,061). These strata are not disjoint. A patient who converts on both outcomes appears in both converter counts, so the four strata sum to 149,602 but cover 147,173 distinct patients. Deterministic hashing (no random seed) makes the cohort exactly reproducible from the universe. Each patient carries $w_i = 1/(\text{stratum sampling fraction})$, equal to 1 for the kept-all strata. Evaluation-time inverse-probability weighting with w_i restores natural prevalence [Horvitz and Thompson, 1952].

Calendar weeks are indexed as $\text{wk} = \lfloor (\text{date} - 2024-01-01)/7 \rfloor$. The relative week is $r = \text{wk} - \text{wk}_{\text{index}}$. The observation horizon is $r \in [0, 52]$ with a fixed absolute window end, so each patient’s observed relative window is $\min(52, 53 - \text{wk}_{\text{index}})$. Late enrollers have shorter windows, handled as right-censoring in the survival likelihood. After a conversion event the patient exits the risk set. Splits are 70/15/15 by patient hash, and no patient appears in more than one split. The dynamics-rich subgroup used in Appendix D is defined as patients with at least five total active exposure weeks (29,859 patients, 20.3%). The originally intended definition of eight or more active weeks is nearly empty under this cohort’s sparsity. It was widened before any model comparison was made on the subgroup.

B Feature definitions

Per-week action vector ($d_a = 17$). $\text{imp_log}(\log(1 + x))$ of the p99-winsorized total weekly impressions), $\log(1 + \cdot)$ of the weekly event count, $\log(1 + \cdot)$ impression volumes in three channel buckets (digital, programmatic, display), three ad-type buckets (display, video, audio), seven targeting buckets (condition, lookalike, behavioral-contextual, conquering, retargeting, geo-demographic, none/other), a week-active flag, and the normalized relative week $r/52$.

Static vector ($d_s = 7$). Source-of-business one-hot (three levels: market-naive, switched from another product, continuing on brand), rx-eligibility flag, dx-eligibility flag, $\log(1 + \cdot)$ prior office-visit count, and normalized enrollment week.

Survival convention. The hazard emitted from state $\mathbf{z}_{i,r}$ predicts an event in week $r+1$. At-risk person-weeks are pooled for training and evaluation. Patients are censored at their observation end.

C Training details

The proposed model is implemented in PyTorch and trained on CPU. The tuned configuration (selected on validation hazard likelihood) uses two GRU layers, hidden size 128, dropout 0.2. The base configuration is one layer, hidden size 64. The one-step task (Table 4) uses the base configuration. The rollout and simulation experiments use the tuned configuration. The static projection W_0 maps the static vector to the initial hidden state of every layer. The hazard head is hidden $\rightarrow$ hidden $\rightarrow$ 1 with ReLU and dropout. The transition head maps $(\mathbf{z}_{i,r}, \mathbf{a}_{i,r})$ through one hidden layer to the 17-dimensional next-week action summary. The loss weight is $\lambda = 0.3$. Optimization: Adam, learning rate 10^{-3} , ReduceLROnPlateau (factor 0.5, patience 2) on validation hazard negative log-likelihood, early stopping (patience 10), at most 50 epochs, batch size 512. A control run at batch size 1024 produced materially identical results, so batch size is not a sensitive choice here. A multi-step rollout training objective (supervising the cumulative incidence along the rollout rather than only one-step hazards) was tried and made calibration worse. It is not used anywhere in this paper. The GRU forecaster ablation sets $\lambda = 0$ and removes the transition head, with nothing else changed. On the rollout task the transition head reaches a held-out skill of +11.5% on both outcomes. Skill is

one minus the ratio of the head’s next-week mean squared error to that of the training-mean predictor, so +11.5% means the head removes 11.5% of the mean predictor’s error. This is a modest figure. It shows that the head has learned part of the exposure dynamics, not that it reproduces individual media plans. In the one-step experiment the same head was also scored against a persistence predictor that carries last week’s exposure forward, and it beats that by 53–54% on both outcomes. The baselines are implemented with scikit-learn (logistic regression, MLP) and LightGBM [Ke et al., 2017] for all GBM variants.

Single training run. Every number in this paper comes from one training run per model and outcome, with no repetition over random seeds. The only estimate of run-to-run variance available is incidental. The operational experiment of Appendix E retrained the dx models, and the retrained world model’s true-future volume error moved by 1.2 to 3.1 percentage points across the four cutoffs relative to Table 5, most visibly from 4.5% to 1.4% at $\tau = 26$. Differences of a few percentage points between coherent models on the dx outcome should therefore be read as ties, and the dx tables below carry no bold for that reason. The rx gaps between the world model and the other coherent models (Table 1) are several times larger than this variance, so they survive the caveat. A multi-seed replication would nonetheless strengthen them and is the first thing a follow-up should do.

D Full results on the diagnosis outcome and subgroups

This appendix reports everything the main text compresses: the one-step foundation task, the full dx trajectory tables, and the dynamics-rich subgroup.

D.1 One-step hazard prediction (T1)

Table 4: One-step hazard prediction (T1), dx outcome, full test split (22,086 patients, 768,431 at-risk person-weeks, 6,856 positives). AUROC is reported on the enriched cohort and IPW-corrected to natural prevalence. AUPRC is on the enriched cohort.

Model	AUROC (enr.)	AUROC (IPW)	AUPRC (enr.)	Cal. slope	ECE
Base rate	0.500	0.500	0.009	—	—
Logistic regression	0.855	0.838	0.051	1.005	0.0011
LightGBM (flat)	0.872	0.855	0.059	0.648	0.0006
MLP	0.873	0.856	0.065	0.929	0.0009
GRU world model (proposed)	0.880	0.864	0.067	0.957	0.0002

On the one-step task the world model is best on every metric, but the margins over LightGBM and the MLP are small, as expected. With a median of three active weeks per patient, cumulative features already capture most of the history, and the one-step task does not exercise the rollout. On the rx outcome (748 test positives) all models land at AUROC 0.84–0.86, within noise of one another. The one-step likelihood cannot separate architectures there, consistent with Section 4.3.

D.2 Trajectory forecasting on dx (T2)

Tables 5–6 and Fig. 3 show the dx outcome in full. The reading is the one predicted in Section 4.4. The structural gap against the per-cell practice is enormous (P1). The three coherent models are within a few percent of each other on volume and ICI, near the $\sim 1\%$ sampling floor of Proposition 2 (P2). Those differences are also within the single-run training variance of Appendix C, so no ordering among the three is claimed on dx. The world model’s patient-level edge over the pooled-hazard GBM is a stable +0.021 to +0.023 AUROC at every cutoff, which is the one dx comparison large enough to read.

D.3 Dynamics-rich subgroup

On patients with at least five total active weeks (20.3% of the cohort), the pooled-hazard GBM degrades to 15–22% volume error while both GRU models stay below 8% at every cutoff, the world

Table 5: Campaign-level trajectory forecasting (T2), dx outcome, test split, full cohort. Volume relative error (%) at the campaign end and coherence. Test risk sets: 20,396 down to 16,106. n_+ = 5,874 down to 1,862. No cell is bolded: the differences among the three coherent models are within the single-run training variance (Appendix C).

Model	Volume rel. error (%) at cutoff τ				Coherent
	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$	
Per- (τ, H) GBM	209.3	256.0	326.7	645.1	54–62%
Per- (τ, H) GBM + future	209.2	255.8	326.6	644.9	27–62%
Naive extrapolation	97.3	96.8	101.9	120.3	100%
Pooled-hazard GBM	8.7	9.2	8.8	6.0	100%
GRU forecaster ($\lambda=0$)	6.6	5.4	3.4	5.7	100%
GRU world model (proposed)	7.5	6.5	4.8	4.5	100%

Table 6: Patient-level discrimination and trajectory calibration on dx (T2), full cohort. AUROC is IPW-corrected. ICI is the mean absolute gap between predicted and KM-observed cumulative-incidence curves. Bold is omitted for the reason given in Table 5.

Model	AUROC (IPW)				Trajectory ICI			
	4	8	13	26	4	8	13	26
Per- (τ, H) GBM	.718	.675	.695	.664	.138	.160	.201	.326
Naive extrapolation	.500	.500	.500	.500	.149	.126	.114	.083
Pooled-hazard GBM	.912	.908	.903	.896	.012	.012	.010	.005
GRU forecaster ($\lambda=0$)	.935	.930	.924	.917	.010	.008	.006	.002
GRU world model (proposed)	.935	.930	.925	.917	.012	.011	.008	.002

model at 1.6–6.8% and the forecaster at 0.3–7.8%. The world model’s trajectory ICI is $2\text{--}3\times$ lower than the GBM’s and its patient-level AUROC edge is up to $+0.02$ (Table 7, Fig. 4). Where there is a sequence worth reading, a learned state beats a summary of it. Between the two GRU models the subgroup comparison is mixed. The forecaster is lower at three of four cutoffs, but these cells have n_+ between 214 and 599, so their sampling floors are 4–7% and the differences are within noise. This is consistent with the main-text finding that transition supervision separates the two models only in the rare-event regime.

E Operational rollout details

This appendix supports Section 6.3. It reports the full operational comparison on both outcomes and the mechanism of the dx degradation. Every method forecasts the campaign’s final volume from cutoff τ without access to the recorded future, except the two “true future” reference rows, which are the retrospective setting of Table 1. The world model rolls out on its own imagined exposures. Each baseline is handed one of three assumed plans: all future exposure off (*zero*), the last observed week held constant (*persist*), or the train-cohort average plan by relative week (*mean plan*). The hybrid control feeds the world model’s imagined exposures into the pooled-hazard GBM hazard head. It is diagnostic for the rare-outcome regime and is reported for rx only.

This experiment trains its own models. For rx it reuses the headline checkpoint, so its true-future rows match Table 1 exactly. For dx it retrains, so its true-future rows differ from the trajectory experiment of Table 5 by ordinary training variance, most visibly 1.4% here against 4.5% there at $\tau = 26$. The closed-loop degradation is therefore read against this experiment’s own true-future reference, not across runs.

On rx (Table 8) the closed-loop world model matches the same model given the true future and beats every baseline, including the baselines given the true future. Its trajectory ICI under closed-loop rollout is 0.0008, 0.0005, 0.0006, 0.0005 at $\tau = 4$ to 26, matching the known-future values 0.0008, 0.0005, 0.0005, 0.0005 to within 0.0001, so imagination preserves whole-curve calibration and not just the endpoint.

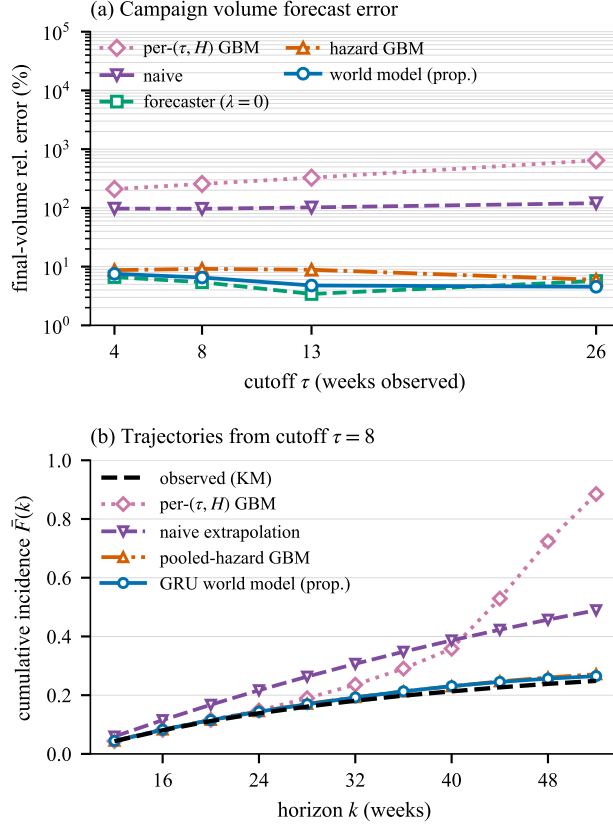


Figure 3: Trajectory forecasting on dx (test split). (a) Final-volume relative error versus cutoff (log scale). (b) Mean predicted cumulative-incidence trajectories from $\tau = 8$ against the Kaplan–Meier curve. The coherent models sit in the same single-digit band. The per-cell practice explodes with the horizon.

On dx (Table 9) the picture reverses. The closed-loop world model degrades to 14.6–26.7%, while the same model given the true future stays at 1.4–8.7% and the pooled-hazard GBM given the true future stays at 6.0–9.2%. The same model under the train-cohort mean plan is unstable, at 4.6%, 51.9%, 115.0%, and 34.7% across the four cutoffs. On dx, then, the world model’s forecast depends strongly on which future it is fed, and no assumed or imagined plan recovers the true-future accuracy.

The mechanism is not settled by these runs, and two readings are consistent with the data. The first is error accumulation through the rollout, the failure mode known from closed-loop sequence models [Ross et al., 2011, Bengio et al., 2015, Janner et al., 2019]. Each imagined week feeds the next, and on dx the hazard head depends on exposure enough for the accumulated error to matter, whereas on rx it does not. The second reading, which we find more likely, is that the sensitivity is to the presence of exposure rather than to its magnitude. Two facts point this way. The dose scan of Table 3 shows that scaling all future exposure from $0.25\times$ to $2\times$ moves the dx forecast by less than 0.003, while switching it off moves the forecast from 0.31 to 0.89. The model has learned that the absence of exposure signals imminent conversion (Section 6.4). An imagined exposure sequence is a sequence of conditional means. Its week-active flag and impression counts sit between the observed values, which are mostly exactly zero or clearly positive, so the state is fed inputs partway toward the pattern that the model reads as exit from the risk set. On rx this channel is weak, since the rx hazard is low and only loosely tied to the exposure pattern, and the imagined rollout stays close to the true-future one. On dx the channel is strong enough to move the campaign-level forecast by tens of percent. The fact that the treatment of the imagined week-active flag had to be selected per outcome (see below) is consistent with this reading. A direct test would hold the imagined activity pattern at the rate observed in the prefix and imagine only the magnitudes. We leave it to future

Table 7: The regime map (volume rel. error, %): the three coherent models on the full dx cohort, the dynamics-rich dx subgroup (≥ 5 total active weeks), and the full rx cohort. Small- n_+ cells are noisy. Bold is omitted. On dx the differences among the three models are within single-run training variance. On rx the ranking is that of Table 1.

Regime	Model	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$
dx, full cohort (n_+ : 5874–1862)	Pooled-hazard GBM	8.7	9.2	8.8	6.0
	GRU forecaster	6.6	5.4	3.4	5.7
	GRU world model	7.5	6.5	4.8	4.5
dx, dynamics-rich (n_+ : 599–214)	Pooled-hazard GBM	15.0	15.2	19.5	21.5
	GRU forecaster	2.3	0.7	0.3	7.8
	GRU world model	4.0	1.6	3.2	6.8
rx, full cohort (n_+ : 711–206)	Pooled-hazard GBM	13.6	17.3	19.2	33.1
	GRU forecaster	5.2	7.4	8.2	11.1
	GRU world model	2.9	2.3	2.6	0.8

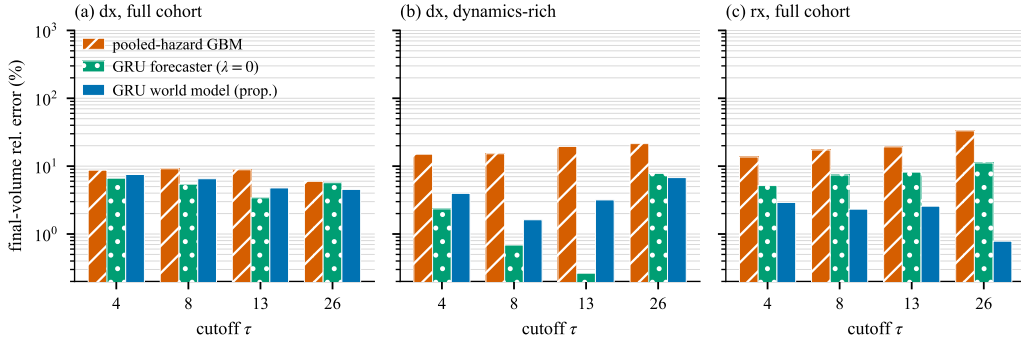


Figure 4: The regime map (Table 7, drawn): final-volume relative error of the three coherent models, log scale. (a) Full sparse dx cohort: the aggregate ceiling of P2. (b) Dynamics-rich patients and (c) the rare rx outcome: the regime dependence of P3.

work. Under either reading the conclusion for deployment is the same. The closed-loop capability is established on the rare outcome, and a deployment on a denser outcome should feed a coherent model an explicit plan whose activity pattern matches the observed data.

Implementation. A single world-model checkpoint per outcome is trained once and reused across cutoffs. For weeks after τ the transition head generates the next exposure, which is fed back into the sequence module with no teacher forcing. The exogenous relative-week index keeps its true value. Imagined actions are clipped per dimension to the training range $[0, p_{99}]$. The week-active flag is fed as a continuous value or thresholded at 0.5, selected on the validation split (rx uses the threshold, dx uses the continuous value). This choice matters on dx and not on rx, which is the pattern the presence-sensitivity reading above predicts.

F Scenario-simulation details (T3)

The simulator is the tuned world model (45 epochs, otherwise as in Appendix C), applied to the full test risk sets at $\tau \in \{4, 8\}$ on the dx outcome. The perturbed pooled-hazard GBM serves as the reference simulator. Dose scaling multiplies all future exposure magnitudes by α . At $\alpha = 0$ this also zeroes the week-active flag, which is what triggers the selection artifact. Per-lever shutoffs zero one action dimension’s future values (13 levers: channel, ad-type, and targeting buckets). The largest observed shift was $+0.0026$ (total digital volume off), and the placebo lever (a negligible bucket scaled $4\times$) moved the forecast by less than 10^{-4} . Within the observed support ($\alpha \in [0.25, 2]$) the mean predicted conversion of the world model varies by less than 0.003 at both cutoffs (Table 3), so the dose-response is flat within support rather than monotone. The rise of about $+0.6$ percentage

Table 8: Operational forecasting, rx outcome, test split, full cohort. Final-volume relative error (%) at cutoff τ . All curves coherent. Test risk sets 21,963 down to 20,595. Bold is omitted: the three world-model rows are all inside the sampling floor of 3.7–7.0%.

Method	Future source	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$
Naive extrapolation	—	64.4	82.6	97.4	166.1
Pooled-hazard GBM	zero	50.6	46.8	35.4	36.3
Pooled-hazard GBM	persist	75.5	83.2	68.5	84.1
Pooled-hazard GBM	mean plan	37.5	39.2	38.7	52.3
Pooled-hazard GBM	true future	13.6	17.3	19.2	33.1
Pooled-hazard GBM	WM imagined (hybrid)	63.5	61.9	58.8	76.5
GRU forecaster ($\lambda=0$)	zero	79.4	65.1	45.7	21.9
GRU forecaster ($\lambda=0$)	persist	75.6	64.3	45.7	21.6
GRU forecaster ($\lambda=0$)	mean plan	64.2	53.5	38.6	20.2
GRU world model	mean plan	2.4	3.6	1.2	3.3
GRU world model	true future	2.9	2.3	2.6	0.8
GRU world model	imagined (closed loop)	2.5	1.7	0.5	1.8

Table 9: Operational forecasting, dx outcome, test split, full cohort. Final-volume relative error (%) at cutoff τ . All curves coherent. Test risk sets 20,396 down to 16,106. Bold is omitted; see text for why the mean-plan row of the world model should not be read as a win.

Method	Future source	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$
Naive extrapolation	—	97.3	96.8	101.9	120.3
Pooled-hazard GBM	zero	119.3	102.8	77.1	26.2
Pooled-hazard GBM	persist	126.3	113.7	85.5	31.3
Pooled-hazard GBM	mean plan	13.5	10.7	8.2	22.0
Pooled-hazard GBM	true future	8.7	9.2	8.8	6.0
GRU forecaster ($\lambda=0$)	zero	206.1	177.4	123.9	31.2
GRU forecaster ($\lambda=0$)	persist	180.3	176.9	123.7	31.1
GRU forecaster ($\lambda=0$)	mean plan	149.8	153.3	117.0	30.2
GRU world model	mean plan	4.6	51.9	115.0	34.7
GRU world model	true future	8.7	7.8	6.6	1.4
GRU world model	imagined (closed loop)	18.8	22.4	26.7	14.6

points occurs only at $\alpha = 4$, outside the support, and should not be quoted as a within-support effect. Per-patient dose-response monotonicity over all α including 0 holds for 20–29% of patients, dragged down by the $\alpha = 0$ artifact. Two protocol refinements are noted for future runs: hold the week-active pattern fixed at $\alpha = 0$ so that only magnitude varies, and restrict reported curves to the observed dose support (approximately $\alpha \in [0.25, 2]$).

G Proofs

Proof of Proposition 1. Part (i) is immediate, because every factor of the product in (8) lies in $[0, 1]$. For part (ii), write the survival functions $\hat{S}(k) = \prod_{m=\tau+1}^k (1 - \hat{h}_{i,m})$ and $S(k) = \prod_{m=\tau+1}^k (1 - h_{i,m})$, and set $P_m^- = \prod_{j=\tau+1}^{m-1} (1 - \hat{h}_{i,j})$ and $P_m^+ = \prod_{j=m+1}^k (1 - h_{i,j})$. Telescoping,

$$\hat{S}(k) - S(k) = \sum_{m=\tau+1}^k P_m^- (h_{i,m} - \hat{h}_{i,m}) P_m^+.$$

Every P_m^-, P_m^+ lies in $[0, 1]$, so the triangle inequality gives (10), and $|\hat{F} - F| = |\hat{S} - S|$. $\square$

Proof of Proposition 2. Unbiasedness follows from assumption (ii) and linearity. The variables $w_i(\mathbf{1}\{\tau < T_i \leq R\} - q_i)$ are conditionally independent by assumption (i), zero-mean, bounded by $w_{\max}$, with variance $w_i^2 q_i(1 - q_i)$. Inequality (11) is Bernstein’s inequality applied to their sum. $\square$

Remark 1 (The $1/\sqrt{n_+}$ floor) *The main text reads volume errors against a relative 1σ floor of $1/\sqrt{n_+}$. Under the assumptions of Proposition 2 the conditional standard deviation of N is $\sqrt{V}$ with $V = \sum_i w_i^2 q_i(1 - q_i)$. Since $w_i \geq 1$ for every patient (Appendix A) and the q_i are small, $V \geq \sum_i w_i q_i(1 - q_i) \approx \mathbb{E}[N | \mathcal{F}_\tau]$, so the relative floor $\sqrt{V}/\mathbb{E}[N | \mathcal{F}_\tau]$ is at least $1/\sqrt{\mathbb{E}[N | \mathcal{F}_\tau]}$. The main text substitutes the observed n_+ for its expectation. The quoted floor is therefore a lower bound on the true floor, and a model whose error sits inside it sits inside the true floor as well. For the rx outcome every converter belongs to a kept-all stratum with $w_i = 1$, so N is the raw converter count.*

H Fisher-information argument

This appendix supports Section 4.3. Fix any encoder parameter u and consider the expected curvature of the two loss terms at an at-risk week. For the hazard term, with logit $\eta_{i,r} = g_{\mathcal{R}}(\mathbf{z}_{i,r})$, the Fisher information contribution is

$$\mathcal{I}_{i,r}^{\mathcal{R}}(u) = h_{i,r+1}(1 - h_{i,r+1}) \left(\frac{\partial \eta_{i,r}}{\partial u} \right)^2 \leq \bar{h} \sup_{i,r} \left(\frac{\partial \eta_{i,r}}{\partial u} \right)^2. \quad (12)$$

The factor $h(1 - h)$ is small because events are rare. Under a Gaussian working model for the transition residuals with variance σ^2 , the transition term contributes

$$\mathcal{I}_{i,r}^{\mathcal{T}}(u) = \frac{\lambda}{\sigma^2} \left\| \frac{\partial \hat{\mathbf{a}}_{i,r+1}}{\partial u} \right\|_2^2, \quad (13)$$

which does not depend on $\bar{h}$ and appears at every observed week, across all d_a components. In our data $\bar{h} \approx 9 \times 10^{-3}$ for the diagnosis outcome and an order of magnitude smaller for the prescription outcome. The rarity factor $h(1 - h) \approx \bar{h}$ therefore throttles the outcome term’s curvature by about two orders of magnitude, while the transition term (13) carries no such factor. This is a scaling argument, not an exact ratio, since the two terms differ in their gradients, in σ^2 , and in λ .

Two caveats apply. First, “two orders of magnitude” is relative to a balanced outcome, for which $h(1 - h)$ would be near $1/4$. With $\bar{h} \approx 10^{-3}$ on rx the factor is about $250\times$, and with $\bar{h} \approx 9 \times 10^{-3}$ on dx about $30\times$. Second, curvature is not usefulness. The argument shows that the transition term supplies most of the information that pins down the shared state when events are rare. It does not show that the directions it pins down are the ones the hazard head needs. That is an empirical question, and the one-flag ablation of Section 6.2 is its test.

I Deployment notes

Three observations from this study bear on how in-flight measurement is set up in practice.

Rare endpoints. On the rx outcome the difference between the world model and the other methods is not incremental. The per-horizon practice produces no usable volume forecast, the coherent GBM baseline is off by 14–33%, and the world model is at the sampling floor. When the endpoint that matters is rare, the sequence model is worth its cost.

Common endpoints. On the dx outcome most of the gap between the per-horizon practice and the world model is closed by the survival formulation alone. Any pooled-hazard model can carry the rollout, including a GBM a team already has. The Fisher argument of Section 4.3 gives a way to judge in advance, from the event rate, whether the sequence model is likely to add anything.

Scenario outputs. The simulator reports what the model expects under a stated plan, coherently and per patient. On targeting-driven data its off-support scenarios measure selection rather than response, and the 0.31-to-0.89 inversion of Section 6.4 is the concrete example. Scenario outputs should carry a support flag and be run with a negative control. Effect claims need experimental design.